

MCA 2003 – EXAMINATION PAPER

1. The simplified value of $\frac{(2002)(2004) - (2003)^2}{(2002)^2 - (2001)(2003)}$ is
 1) 1 2) -1
 3) 2003 4) $1/2003$

2. Kalpana has certain number of flowers. It is not less than 15 and not more than 30. If she wants to give these flowers equally to two of her friends, then one flower remains. If she wants to give these flowers equally to three of her friends, then two flowers remain. If she wants to give these flowers equally to four of her friends, then three flowers remain. The number of flowers with Kalpana is
 1) 11 2) 23
 3) 25 4) 35

3. Let x and y be the largest and smallest 4 digit numbers formed by 6, 1, 7 and 4 using each digit only once in each x and y. The number x - y is
 1) divisible by 9 and not by 7
 2) divisible by 7 and not by 9
 3) divisible by both 9 and 7
 4) divisible neither by 9 nor by 7

4. Last year my age was 5 times my daughter's age. Next year, my age will be only 4 times my daughter's age. Now my daughter's age is
 1) 3 years 2) 5 years
 3) 6 years 4) 7 years

5. The least positive integer which can be expressed as the sum of two distinct non-zero perfect squares in two different ways is
 1) 25 2) 65
 3) 625 4) 1729

6. In Co-optex, 40% discount is given for handloom sarees. After deducting the discount, for the balance amount, further discount of 20% is also given. Effectively, the total discount on the original price is
 1) 60% 2) 58%
 3) 52% 4) 48%

7. At 6 a.m. the hour hand and the minute hand are just opposite to each other. What is the next immediate time at which they will be opposite to each other?
 1) 6 hrs. $5\frac{5}{11}$ min. a.m.
 2) 7 hrs. 5 min. a.m.
 3) 7 hrs. $5\frac{5}{11}$ min. a.m.
 4) 6 p.m.

8. Praveen wanted to read a story book which contains 300 pages. He read 1 page on the first day, 2 pages on the second day, 3 pages on the third day and so on. How many days it took for Praveen to complete the book?
 1) 20 days 2) 24 days
 3) 25 days 4) 30 days

9. A man drove at a speed of 50 km/hr. to reach Madurai from Trichy. Because of bad weather, he drove back to Trichy only at a speed of 30 km/hr. The average speed for the entire trip is
 1) 40 km/hr.
 2) 42.5 km/hr.
 3) 37.5 km/hr.
 4) indeterminate because of insufficient data

10. In a chess board, eight rooks are to be placed such that no two rooks are in the same row or in the same column. In how many ways the eight rooks can be placed in the chess board?
 1) 64 2) 5040
 3) 40320 4) 16777216

Directions for questions 11 and 12 :

My husband purchased 3 different types of fruits namely, banana, mango and apple. I asked him how many fruits did he purchase in each kind. He replied that the number of bananas exceeds one-fifth of the total number of fruits by one. The number of mangoes exceeds one-fourth of the total number of fruits by one. The number of apples exceeds half of the total number of fruits by one.

11. How many apples did my husband purchase?

1) 13 2) 16
3) 31 4) 43

12. If the cost of one banana is Rs. 2, one mango is Rs. 9 and one apple is Rs. 10, how much did he pay for all the fruits?

1) Rs. 336 2) Rs. 420
3) Rs. 480 4) Rs. 496

13. If the height of a cone is increased by 25% and the radius is decreased by 20%, then the volume of the cone is

1) reduced by 20% 2) reduced by 15%
3) increased by 15% 4) increased by 20%

14. How many numbers are there between 1 and 100 which are either divisible by 7 or containing the digit 7?

1) 14 2) 30
3) 32 4) 34

15. If Rs. 30,000 is deposited for a period of three years at the rate of 7% per annum and if the interest is accumulated to the capital every one year, what is the interest after three years nearest to a multiple of ten rupees?

1) Rs. 6,300 2) Rs. 6,750
3) Rs. 6,760 4) Rs. 7,200

16. If 60 percent of 60 percent of a number is 60, what is 40 percent of 40 percent of that number?

1) 40 2) 80
3) 135 4) 80/3

17. The smallest four digit prime number is

1) 1001 2) 1003
3) 1007 4) 1009

Directions for questions 18 - 20 :

A software project can be completed by Priya in 40 days. If Bama joins with Priya, then they can complete the project in 24 days. If Hema joins with Priya and Bama, then they can complete the project in 15 days. If Kavitha also joins with these three girls, then the project can be completed just in 9 days.

18. Among these four girls, which pair has equal efficiency?

1) Priya and Hema 2) Kavitha and Bama
3) Bama and Hema 4) Priya and Kavitha

19. Which girl is more efficient?

1) Priya 2) Bama
3) Hema 4) Kavitha

20. If Bama alone does the project, in how many days will she complete the project?

1) 60 days 2) 45 days
3) 40 days 4) 30 days

21. Arun sold three old almirahs, each for Rs. 600, one for a loss of 40% and the other two for a profit of 20% and 25% respectively. The net profit or loss for Arun is

1) 5% profit 2) 9% profit
3) $9\frac{1}{11}\%$ loss 4) 10% loss

22. If 10 machines can produce 10 thousand identical items in 10 days by operating for 10 hrs./day, how many days does it take for 5 machines to produce 5 thousand such items by operating for 5 hrs./day?

1) 5 days 2) 10 days
3) 20 days 4) 40 days

23. In a class with boys and girls, there are 40 students and their average weight is 46.9 kg. The average weight of boys is 47.6 kg. and the average weight of girls is 44.1 kg. The number of girls in the class is

1) 8 2) 10
3) 12 4) 15

24. If the sum of two positive numbers is 30, the maximum product of these two numbers is

- 1) 200 2) 225
3) 300 4) 125

25. Shoba and her brother have totally 200 books. After getting 25 books from Shoba, her brother finds that he has 75 books more than one-fourth of the number of books that Shoba has now. In the beginning how many books did Shoba have?

- 1) 75 2) 100
3) 125 4) 150

26. The two digit number which exceeds the sum of the squares of its digits by 18 is

- 1) 27 2) 28
3) 42 4) 43

27. In a group of 60 students, 40 students have studied Operations Research and 27 students have studied Graph Theory. Atmost how many of them have not studied either of these two subjects?

- 1) 7 2) 13
3) 20 4) 27

28. ABC is a right angled triangle with $\angle A = 90^\circ$, $AB = 14$ cm and $AC = 48$ cm. The length of the perpendicular from A to BC is

- 1) 11.44 cm 2) 12.44 cm
3) 13.44 cm 4) 14.44 cm

29. If x and y are real numbers and $x > y$, then which one of the following is always true?

- 1) $|x| > |y|$ 2) $x^2 > y^2$
3) $x(x+1) > y(y+1)$ 4) $2y-1 < 2x-1$

30. The perimeter of a rectangle is 20 metres. Its area remains the same when its length is increased by 2 metres and its breadth is decreased by 1 metre. The area of the rectangle is

- 1) 16 sq.m. 2) 21 sq.m.
3) 24 sq.m. 4) 25 sq.m.

Directions for questions 31 - 34 :

On 1st, 11th and 21st of every month, I save one rupee. The amount of savings is doubled on the next day and the doubling of savings is continued upto 10th, 20th and the last day of the month.

31. The total savings in the month of April will be

- 1) Rs. 3,069 2) Rs. 3,072
3) Rs. 4,093 4) Rs. 6,141

32. The total savings in February, 2003 will be

- 1) Rs. 2,301 2) Rs. 2,557
3) Rs. 4,605 4) Rs. 5,117

33. The excess savings in the month of October over the savings in the month of September will be

- 1) nothing 2) Rs. 512
3) Rs. 1,024 4) Rs. 2,048

34. The total savings for the whole year 2003 nearest to a multiple of thousands of rupees is

- 1) Rs. 36,000 2) Rs. 42,000
3) Rs. 43,000 4) Rs. 50,000

35. The least common multiple of two numbers is 168. If one number is 42, the other number should be a multiple of

- 1) 3 2) 6 3) 7 4) 8

36. Two girls and their boy friends want to take a photograph standing in a row. No girl wants that her boy friend to stand adjacent to the other girl. In how many ways the arrangement can be done?

- 1) 4 2) 6 3) 8 4) 12

37. There are 100 questions in a two hour M.C.A. entrance test. Among these questions, 40 will be in the area of quantitative ability. It is suggested that two and a half times as much time be allowed for each quantitative ability problem as for each of the other questions. How many minutes should be spent on the quantitative ability problem?

- 1) 30 2) 60 3) 75 4) 80

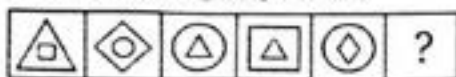
54. How many pairs of letters are there in the word **STRENGTH** which have as many letters between them as in the alphabet?

- 1) one 2) two
3) more than two 4) no such pair

55. If **WHITE** is coded as **5209823**, then **BLACK** will be coded as

- 1) 1131122 2) 1131212
3) 2121311 4) 2211311

56. Which of the figures 1, 2, 3, 4 given in the answers will logically fit into the next figure in the following sequences?



- 1) 2) 3) 4)

Directions for questions 57 - 60 :

In a group of five persons Anita, Beena, Chitra, Dorothy and Emile, one of them is an advocate, one is a doctor, one is an executive, one is a software engineer and the other one is a teacher. Anita, Chitra and the teacher prefer coffee and two of them namely Beena and the executive prefer tea to coffee. The software engineer is Chitra's sister and both of them prefer the same drink. The doctor and Emile prefer to drink coffee.

57. Who is the software engineer?

- 1) Anita 2) Beena
3) Dorothy 4) Emile

58. Who is the doctor?

- 1) Anita 2) Beena
3) Chitra 4) Dorothy

59. Who is the executive?

- 1) Anita 2) Chitra
3) Dorothy 4) Emile

60. Which pair prefer tea?

- 1) Advocate and Dorothy
2) Beena and Emile

- 3) Software engineer and Advocate
4) Anita and Doctor

Directions for questions 61 - 63 :

Three adjacent faces of a cube are painted with blue colour and the other three faces are painted with red colour. The cube is cut into 125 equal small cubes.

61. How many small cubes will have paint on only one face?

- 1) 27 2) 54 3) 36 4) 117

62. How many small cubes will have no paint on any face?

- 1) 27 2) 54 3) 64 4) 117

63. How many small cubes will have the same colour on two faces and no paint on the remaining four faces?

- 1) 12 2) 18 3) 24 4) 36

64. What is the Hexadecimal equivalent of the decimal number 44975?

- 1) AFAF 2) BEBE
3) EBEB 4) FAFA

65. The binary equivalent of the octal number 13.54 is

- 1) 1110.1101 2) 1001.1110
3) 1101.1110 4) 1011.1011

66. If 123 in hexadecimal system is x in octal system, then x is

- 1) 123 2) 291
3) 344 4) 443

67. The number 43 in 2's complement representation is

- 1) 101011 2) 101010
3) 010100 4) 010101

68. The simplified form of the Boolean expression $xyz' + xy'z' + x'yz'$ is

- 1) $xy + x'z' + xz'$ 2) $(y + z')x$
3) $(x + z')y$ 4) $(x + y)z'$

69. In a shop, the price of an item is Rs. 3,455. It is coded as GLOO. Another item whose price is Rs. 580 is coded as OLY. What is the code of price Rs. 3,085?
 1) IGLO (2) GYLO 3) ILOY 4) GOLY
70. A is the daughter of E, B is the son of C, D is the husband of C and E is the daughter of D. Then how B is related to A?
 1) Brother 2) Father
 3) Husband 4) Uncle
71. ROM is composed of
 1) floppy disks 2) magnetic cores
 3) microprocessors 4) photoelectric cells
72. A micro computer has primary memory of 640K. What is the exact number of bytes contained in this memory?
 1) 640,000 (2) 655,360
 3) 65,536 4) 64×2^{16}
73. Find the odd man out
 1) coaxial cable 2) optical fibre
 3) twisted pair wire 4) microwaves
74. Which one of the following when added can reduce the processing time of a computer?
 1) a buffer 2) a converter
 3) a co-processor 4) an EPROM
75. What name is given to the collection of facts, items of information or data which are related in some way?
 1) information tree 2) information provider
 3) directory information 4) database
76. Who is considered to be the father of Artificial Intelligence?
 1) George Boole 2) John McCarthy
 3) Allen Newell 4) Alan Turing
77. Which command combines the contents of one file with another?
 1) APPEND 2) ADD
 3) RENAME 4) RESTORE
78. Choose the operator that you cannot overload in C++
 1) = =
 3) ! =
 (2) > =
 (4) ? :
79. Find the odd man out
 1) altavista 2) google
 3) java 4) lycos
80. Consider the following recursive Pascal function where n is a non-negative integer :
 function calc (n:nonnegint) : nonnegint; begin
 if n = 0 then
 calc := 0
 else
 calc := n + calc (n - 1)
 end;
 What will be the value returned by calc when it is invoked with n = 10?
 1) 10 2) 45 (3) 55 4) 90
81. The temperature of 32° C is equal to
 1) 0° F (2) 89.6° F
 3) 100° F 4) 115.2° F
82. A gallon is approximately equal to
 1) 3.5 litres 2) 4.0 litres
 3) 4.55 litres 4) 6.25 litres
83. Who is the inventor of telephone?
 1) Alexander Graham Bell
 2) J. L. Baird
 3) Thomas Alva Edison
 4) Marconi
84. If a body is allowed to fall from the window of a moving train, the body will fall to the ground after moving along
 1) a parabolic path 2) a straight line path
 3) a circular path 4) an elliptic path
85. Numismatics is the study of
 1) numbers
 2) coins and medals
 3) weights and measures
 4) calculators

86. Which of the following is/are true about haemoglobin?

- a) It is an oxygen carrying substance in blood
b) It is a macro molecule
c) It is rich in iron
d) It gives red colour to blood
1) (a) and (d) only 2) (c) and (d) only
3) (b), (c) and (d) only 4) all (a), (b), (c) and (d)

87. Euclid is associated with

- 1) geometry 2) atomic theory
3) theory of relativity 4) law of heredity

88. The instrument for measuring the relative humidity of the atmosphere is

- 1) hygrometer 2) hydrometer
3) lactometer 4) altimeter

89. The centre of gravity of a solid right circular cone of height h lies on the axis of the cone at a distance d from its vertex where $d =$

- 1) $h/3$ 2) $2h/3$ 3) $h/4$ 4) $3h/4$

90. The brightest planet is

- 1) Jupiter 2) Saturn 3) Mercury 4) Venus

Directions for questions 91 and 92 :

Each of the following words in capital letters is followed by four likely synonyms. One of them is NOT the correct synonym. Pick up that.

91. CAPRICIOUS :

- 1) unpredictable 2) wavering
3) fickle 4) approachable

92. TOLERATE :

- 1) abide with 2) bear 3) endure 4) repine

Directions for questions 93 - 95 :

Each of the question below consists of a word in capital letters followed by four words. Choose the one that is most nearly similar in meaning to the word in capital letters.

93. ABORTIVE :

- 1) successful 2) fruitless
3) insulting 4) moderate

94. FRANCHISE :

- 1) discount 2) licence
3) enthusiasm 4) sweet smell

95. TAUTOLOGY :

- 1) absurdity 2) tension
3) perseverance 4) repetition

Directions for questions 96 and 97 :

Select the pair of words which are related in the same way as the pair of words in capital letters are related to each other.

96. ARROGANCE : MODESTY

- 1) army : soldier 2) fat : obese
3) strength : debility 4) speak : shout

97. PARTISANSHIP : DISPASSIONATE

- 1) zealousness : enthusiastic
2) sorrow : disconsolate
3) moderation : intemperate
4) involvement : ardent

Directions for questions 98 - 100 :

In each of the following sentences, four words or phrases have been underlined. Only one underlined part in each sentence is NOT acceptable in standard English. Find out the trouble spot.

98. A tiger is the National animal of India
1 2 3 4

99. The journey by car is as quick and perhaps
1 2
quicker than by train.
3 4

100. Long life is good if one be happy and
1 2 3
has friends.
4

ANSWERS

1 2	2 2	3 3	4 4	5 1	6 3	7 3	8 2	9 3	10 3
11 3	12 3	13 1	14 2	15 2	16 4	17 4	18 1	19 4	20 1
21 3	22 3	23 1	24 2	25 3	26 4	27 1	28 3	29 4	30 3
31 1	32 1	33 3	34 3	35 3	36 3	37 3	38 1	39 1	40 1
41 3	42 4	43 2	44 1	45 2	46 1	47 2	48 2	49 4	50 3
51 2	52 2	53 3	54 2	55 1	56 1	57 1	58 3	59 3	60 1
61 2	62 1	63 1	64 1	65 4	66 4	67 1	68 1	69 2	70 4
71 4	72 2	73 4	74 3	75 4	76 2	77 1	78 4	79 3	80 3
81 2	82 3	83 1	84 3	85 2	86 4	87 1	88 1	89 4	90 4
91 4	92 4	93 2	94 2	95 4	96 3	97 3	98 1	99 2	100 3